

- 7 (a) The variation with time t of the current I for a sinusoidally alternating voltage supply of peak value of 240 V is shown in Fig. 7.1.

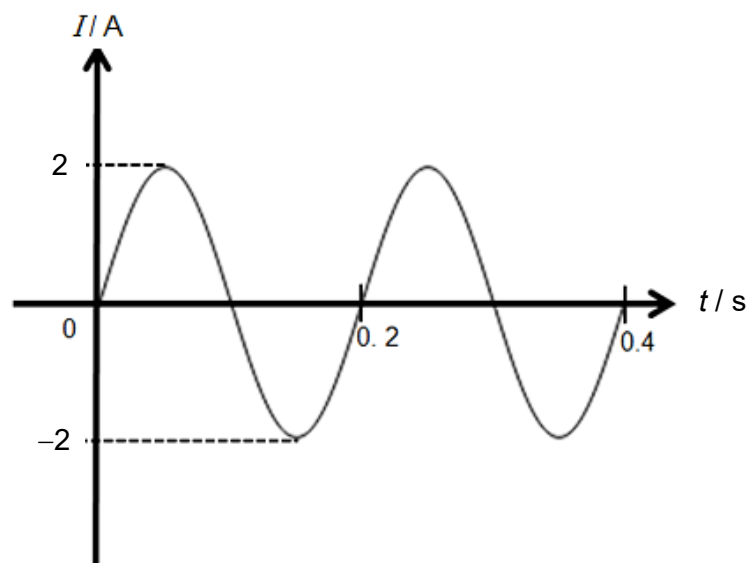


Fig. 7.1

- (i) Determine the root-mean-square value of the current.

current = A [2]

- (ii) Determine the mean power supplied.

mean power = W [2]

(b) A transformer has 500 turns in the primary coil and 10 turns in the secondary coil. The alternating voltage supply in (a) is connected to the primary coil.

(i) Determine the peak values of the voltage and current in the secondary coil.

State an assumption made.

voltage in secondary coil = V

current in secondary coil = A [2]

Assumption:

..... [1]

(ii) Suggest one use for such a current from the secondary coil of the transformer.

.....

..... [1]

Section B

Answer **one** question from this Section in the spaces provided.

- 8 (a) A horizontal string is stretched between two fixed points A and B. A vibrator is used to